MILITARY SPECIFICATION

PROJECTILE, 60 MM ILLUMINATING, XM721, METAL PARTS FOR

This Specification is approved for use within the U.S. Armament, Munitions and Chemical Command, and is available for by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for metal parts and sub-assemblies for the projectile, 60mm, Illuminating, XM721.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards from a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1005 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing, an Marking for Shipment of Inert Ammunition Components

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 1310

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9345340 - Projectile, 60mm Illuminating, XM721 Loading Assembly

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contacting activity).

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Method
E-8 - Tension Testing of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philidelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed amomg technical groups and using Federal Agencies).

2.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings, specifications and this specification.

3.2 Components and assemblies. The components and assemblies shall comply with the applicable requirements specified on drawing (dwg.) 9345340 and associated drawings and with all requirements specified in applicable specifications, standards, and this specification. No part may have a burr which might interfere with the assembly or function of the round.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. Threads shall be full and undamaged for the entire minimum length or depth. No part may have a burr which might be injurious to personnel handling the item. All parts shall be free of chips or foreign matter. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated with residue.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contacting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative (s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative (s) provide equal assurance, the provisions of this specification shall apply.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contacting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I First article inspection

MIL-P-70715 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter (dwg. 934530)				
	Examination for defects	10		3.2	4.4.2.1
	Cone, Tail (dwg. 9345342)				
	Examination for defects	10		3.2	4.4.2.2
	Plate,Base (dwg. 9390658)				
	Examination for defects	10		3.2	4.4.2.3
	Cage, Bearing (dwg. 9390657)				
	Examination for defects	10		3.2	4.4.2.4
	Pin (dwg. 9381088)				
	Examination for defects	10		3.2	4.4.2.5
	Plate, Parachute, Expulsion (dwg. 9345344)				
	Examination for defects	10		3.2	4.4.2.6
	Protector, Cable (dwg. 9345341)				
	Examination for defects	10		3.2	4.4.2.7
	Swivel Assembly (dwg. 9390656)				
	Examination for defects	10		3.2	4.4.2.8
	Body, Tube (dwg. 9345349).				
	Examination for defects	10		3.2	4.4.2.9
	Body Tube Assembly (dwg 9345348)				
	Examination for defects	10		3.2	4.4.2.10
	Ring, Filler (dwg 9390654)				
	Examination for defects	10		3.2	4.4.2.11

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I First article inspection

MIL-P-70715 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Disc, Ignition (dwg. 9369944)				
	Examination for defects	10		3.2	4.4.2.12
	Plate, Support (dwg. 9369945)				
	Examination for defects	10		3.2	4.4.2.13
	Parachute, Flat Circular (dwg. 9345346)				
	Examination for defects	10		3.2	4.4.2.14
	Breaking strength of nylon tape	10		3.2	
	Breaking strength of suspension line	10		3.2	

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Adapter		9345350
			NEXT HIGHER ASSEMBLY 9345348

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall outside diameter		0.40%	3.2	Gage
102	Forward outside diameter		0.40%	3.2	Gage
103	Rear outside diameter		0.40%	3.2	Gage
104	Diameter of slot		0.40%	3.2	Gage
105	Width of slot		0.40%	3.2	Gage
106	Overall length		0.40%	3.2	Gage
107	Width of flange		0.40%	3.2	Gage
108	Distance to end of flange		0.40%	3.2	Gage
109	Inside diameter		0.40%	3.2	Gage
Minor					
201	Chamfers missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.4	Visual

REMARKS

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Cone, Tail		9345342
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total runout of thread with axis of inside diameter and rear shoulder		0.40%	3.2	Gage
102	Perpendicularity of rear shoulder with axis of inside diameter		0.40%	3.2	Gage
103	Runout of obturator groove with axis of inside diameter		0.40%	3.2	Gage
104	Total runout of bourrelet with axis of inside diameter		0.40%	3.2	Gage
105	Total runout of forward outside diameter with axis of inside diameter		0.40%	3.2	Gage
106	Wall thickness		0.40%	3.2	Gage
107	Obturator groove diameter		0.40%	3.2	Gage
108	Bourrelet diameter		0.40%	3.2	Gage
109	Forward OD		0.40%	3.2	Gage
110	Pitch diameter of rear thread				

REMARKS

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Cone, Tail		9345342
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Major diameter of rear thread, min.		100%	3.2	Gage
112	Overall length		0.40%	3.2	Gage
113	Perpendicularity of forward shoulder with axis of inside diameter		0.40%	3.2	Gage
114	Forward inside diameter		0.40%	3.2	Gage
115	Thru hole missing		0.40%	3.2	Visual
116	Presence of burrs in threads		0.40%	3.2	Visual
117	Width of obturator groove		0.40%	3.2	Gage
118	Total runout of OD with axis of ID at rear		0.40%	3.2	Gage
119	Surface finish of press-fit surface		0.40%	3.2	Gage
120	Diameter at base of I.D.		0.40%	3.2	Gage
121	Full thread, min		0.40%	3.2	Gage
122	Tension test		(a)	3.1	Gage
Minor					
201	Radii incorrect or blends discontinuous		0.65%	3.2	Visual
202	Presence of burrs or foreign matter		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

(a) Two samples shall be selected per material interfix.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Plate, Base	1		9390658
				NEXT HIGHER ASSEMBLY: 9390656

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70715 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Cage, Bearing		9390657
			NEXT HIGHER ASSEMBLY 9390656

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of plate		0.40%	3.2	Gage
102	Diameter of cage		0.40%	3.2	Gage
103	Diameter of well		0.40%	3.2	Gage
104	Diameter of thru-hole		0.40%	3.2	Gage
105	Depth of well		0.40%	3.2	Gage
106	Material damaged		0.40%	3.2	Visual
107	Runout of OD with well		0.40%	3.2	Gage
108	True position of thru-hole with well		0.40%	3.2	Gage
109	Well surface finish improper		0.65%	3.2	Visual
110	Thickness of sidewalls, min		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

REMARKS

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Pin		9381088
			NEXT HIGHER ASSEMBLY 9390656

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter of base of bin		0.40%	3.2	Gage
102	Outside diameter above base		0.40%	3.2	Gage
103	Distance from side to center of hole		0.40%	3.2	Gage
104	Distance from top to center of hole		0.40%	3.2	Gage
105	Diameter of thru-hole		0.40%	3.2	Gage
106	Height of base		0.40%	3.2	Gage
107	Overall height		0.40%	3.2	Visual
108	Material damaged or defective		0.40%	3.2	Gage
109	Diameter of countersink				
Minor					
201	Surface finish improper		0.65%	3.2	Visual
202	Radii incorrect		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70715(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Plate, Parachute, Expulsion		9345344
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Large outside diameter		0.40%	3.2	Gage
102	Small outside diameter		0.40%	3.2	Gage
103	Overall thickness		0.40%	3.2	Gage
104	Pitch diameter		0.40%	3.2	Gage
105	Minor diameter		0.40%	3.2	Gage
Minor					
201	Thickness of base		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Protector, Cable		9345341
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
103	Radius of inside diameter		0.40%	3.2	Gage
104	Radius of outside diameter		0.40%	3.2	Gage
Minor					
201	Angle of circumference		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-70715 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Swivel Assembly				9390656
					NEXT HIGHER ASSEMBLY 12630532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of adhesive on pin or in cage		0.40%	3.2	Visual
102	Pin does not spin freely		0.40%	3.2	Visual/Manual
103	Outside diameter. max		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Body, Tube		9345349
			NEXT HIGHER ASSEMBLY 9345348

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Overall length		0.40%	3.2	Gage
103	Circularity		0.40%	3.2	Gage
104	Perpendicularity of front face with I.D.		0.40%	3.2	Gage
105	Perpendicularity of read face with I.D.		0.40%	3.2	Gage
Minor					
201	Chamfer missing or incomplete		0.65%	3.2	Visual
202	Radius missing or incomplete (adapter end)		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	Body Tube Assembly		9345348
			NEXT HIGHER ASSEMBLY 9345340

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum diameter		0.40%	3.2	Gage
102	Overall length		0.40%	3.2	Gage
103	Pitch diameter of thread		0.40%	3.2	Gage
104	Minor diameter of thread		0.40%	3.2	Gage
105	True position of inside diameter with pitch diameter and adapter face		0.40%	3.2	Gage
106	Pull test		0.40%	3.2	Gage
107	Protective finish missing, damaged		0.40%	3.2	Visual
108	Outside diameter of front face		0.40%	3.2	Gage
109	Runout (2 places) of adapter O.D. with pitch diameter		0.40%	3.2	Gage
Minor					
201	Presence of protective finish on excluded surfaces		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Ring, Filler	1 OF 1	9390654
			NEXT HIGHER ASSEMBLY: 9390653, 12630532

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
Major					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Canister, Illuminant	1 OF 1	9369944
			NEXT HIGHER ASSEMBLY 9390652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
104	Diameter of thru-hole		0.40%	3.2	Gage
Minor					
201	Radius improper		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.13	Plate, Support		9369945
			NEXT HIGHER ASSEMBLY 9390652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Inside diameter		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.4	Visual

REMARKS

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Parachute, Flat Circular				9345346
					NEXT HIGHER ASSEMBLY 9345345
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole or tear in canopy		0.40%	3.2	Visual
102	One or more shroud lines missing		0.40%	3.2	Visual
103	Stitching missing or broken		0.40%	3.2	Visual
104	Shroud line ends tape missing		0.40%	3.2	Visual
105	Diameter of vent hole		0.40%	3.2	Gage
106	Missing reinforcing tape		0.40%	3.2	Visual
107	Distance from edge of parachute to end of suspension line		0.40%	3.2	Gage
108	Parachute O.D.		0.40%	3.2	Gage
Minor					
201	Length of reinforcing tape, min		0.65%	3.2	Visual
202	Distance between pressure sensitive tape segments		0.65%	3.2	Visual
203	Overlap of edge improper		0.65%	3.2	Visual
204	Overlap at center improper		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.4	Visual
NOTES:					

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" Column in the table starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and Processes. Compliance with all Requirements of Section 3 of this specifications shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 of 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Tension test. From each of two tail cones selected from one interfix, one tension test specimen of the longest obtainable round or flat (Standard or subsized) specimen conforming to ASTM E-8 shall be taken longitudinally from the tapered portion of the component. Failure of any specimen to meet applicable requirements shall be cause for rejection of the applicable material interfix. Equipment and test procedures shall be as prescribed by ASTM E-8 except that elongation shall be measured to the nearest thousandth (0.000) prior to Conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0). Failure of any specimen to achieve the required tensile strength or elongation shall be cause for rejection of all material from the same mill heat of steel. Actual test by an independent source (not only mill certification) must be performed.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with the requirements of Level C of MIL-STD-1169.

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6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly to XM721 illuminating cartridges for use on the 60mm mortar.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army ARDEC, ATTN: AMSMC-QAT-X(D), Bldg 62, Picatinny Arsenal, NJ 07806-500. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading of US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal Picatinny Arsenal or ARRADCOM. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Canister
Cartridge
Illuminant
Projectile

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1310-A432)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70715	PROJECTILE, 60MM ILLUMINATING, XM721 METAL PARTS FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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